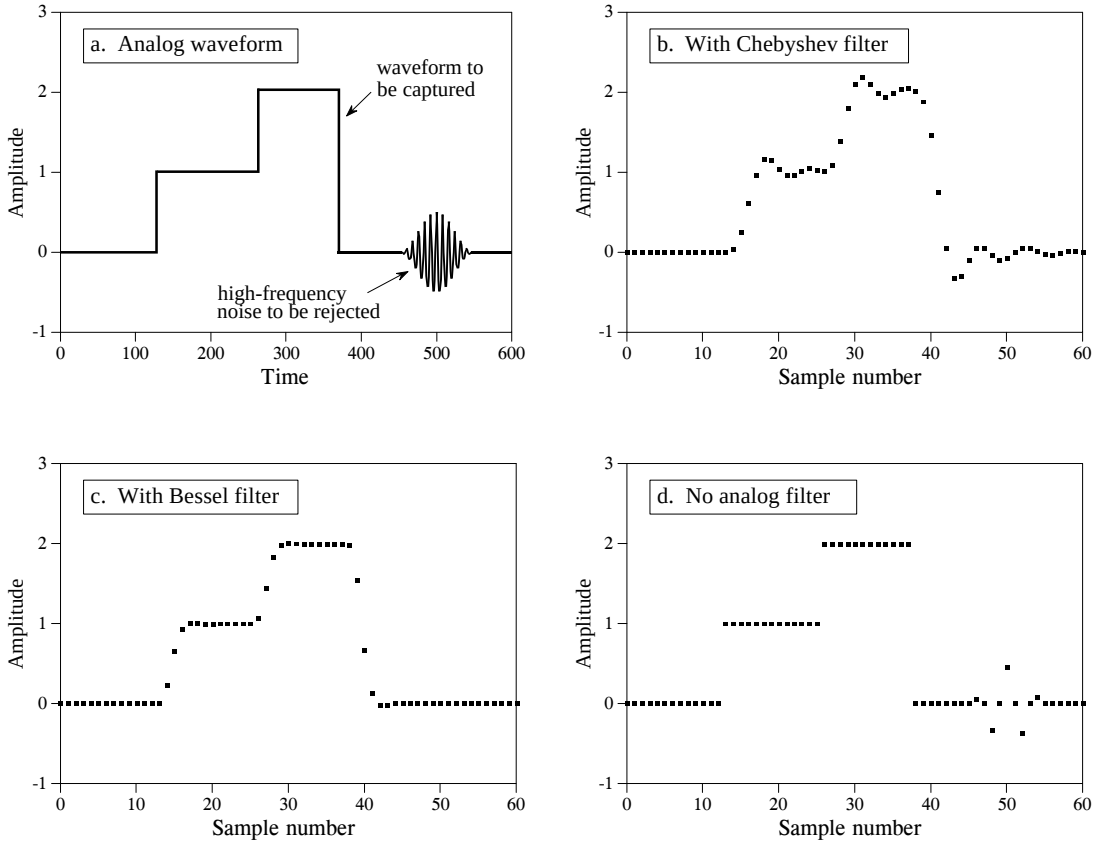

FIGURE 3-15
Three antialias filter options for time domain encoded signals. The goal is to eliminate high frequencies (that will alias during sampling), while simultaneously retaining edge sharpness (that carries information). Figure (a) shows an example analog signal containing both sharp edges and a high frequency noise burst. Figure (b) shows the digitized signal using a *Chebyshev filter*. While the high frequencies have been effectively removed, the edges have been grossly distorted. This is usually a terrible solution. The *Bessel filter*, shown in (c), provides a gentle edge smoothing while removing the high frequencies. Figure (d) shows the digitized signal using *no antialias filter*. In this case, the edges have retained perfect sharpness; however, the high frequency burst has aliased into several meaningless samples.

There are many useful benefits in sampling *faster* than this direct analysis. For example, imagine redesigning the digital voice recorder using a 64 kHz sampling rate. The antialias filter now has an easier task: pass all freq-uencies below 3 kHz, while rejecting all frequencies above 32 kHz. A similar simplification occurs for the reconstruction filter. In short, the higher sampling rate allows the eight pole filters to be replaced with simple resistor-capacitor (RC) networks. The problem is, the digital system is now swamped with data from the higher sampling rate.

The next level of sophistication involves **multirate** techniques, using more than one sampling rate in the same system. It works like this for the digital voice recorder example. First, pass the voice signal through a simple RC low-